ESSAY · A DISPUTATION, MARSHALL MCLUHAN PRESIDING

Whether the Cloud Makes Many Worlds

A disputation in the old manner, held in a Toronto hall, with Marshall McLuhan presiding as master and the dead on the benches. The question: whether the digital cloud has made of us many worlds. The determination: not many worlds, but no world; and the bonds were never cut.

Richard Verver · October 2026

> *Take these evidences, this evidence of the purchase, both which is sealed, and this evidence which is open; and put them in an earthen vessel, that they may continue many days.*
> — Jeremiah 32:14

> *And the light shineth in darkness; and the darkness comprehended it not.*
> — John 1:5

A pastiche. In the universities of the thirteenth century a master proposed a question; opponents argued against the answer he meant to give; a bachelor replied as best he could; and the master then *determined* the question and answered each objection in turn. What follows is a *reportatio*, a student's record of what was said in the hall. Words quoted as any speaker's are his or her own, from the works named in the note at the end. Everything else they say is invented. Facts about the present are sourced in that note; the verdicts are the speakers'. Marshall McLuhan (1911–1980) taught English at St Michael's College in the University of Toronto for most of his career, and was received into the Catholic Church in 1937.

I. The question proposed

The hall at St Michael's College, Toronto, on the feast of St Teresa of Ávila. The benches are full, and most of those on them have been dead for some time. By the door the beadle has set one chair more than the company needs. Nobody sits in it.

The bell rang twice. The master came in late, a gown thrown over a tweed jacket and a newspaper folded under his arm, and climbed into the chair.

McLuhan. Sit down, all of you. Those who can't sit, hover. *(Laughter.)* The last time this house met, a Scotsman proved that the price of eggs had been made private. He was right, but the price was only the first thing. Since then the news has been made private, and the past, and the pictures of the past. So here is the question, put as the old masters put theirs, so that it can be answered yes or no: *whether the cloud makes of us many worlds.*

By the cloud I mean what the young mean: the warm fog of servers and screens through which everything now reaches them, one at a time, adjusted to each. You are all in it, even the dead. Especially the dead. Most of you are in it more than you were ever in print.

My bachelor will answer. No one knows his name, which in this house is a qualification. He wrote a little book in the fourteenth century about a cloud, so he has the advantage of us. *(He nods to a thin man in a grey habit at the lectern below.)* The opponents may begin.

A student *(near the door)*. Master, the chair by the door. Who is it for?

McLuhan. For a guest who is always expected. Leave it empty.

II. It seems that the cloud makes many worlds

The first opponent to rise was a small man in a full-bottomed wig, very polite and very sure of himself.

Leibniz. It seems, Master, that it does. I proved long ago that every simple substance, every soul, has "no windows." Nothing comes in and nothing goes out. Each mirrors the whole universe from its own point of view, and they keep in step only because God established their harmony before any of them existed. Your cloud has perfected my system. It gives each soul its own view and keeps them all in step from a server farm. I am told that a company which rents out films shows each subscriber a different picture on the front of the same film: the lovers to one who likes love stories, the comedian to one who likes comedies. As I wrote, "the same city regarded from different sides appears entirely different." Each subscriber, then, a world.

McLuhan. Finish the sentence.

Leibniz. Later, Master. *(Laughter.)*

The second was a thin old gentleman from Königsberg, who had never before been so far from home.

Kant. It seems further that it does. We never know things as they are in themselves, only as they appear to us under the forms we bring: space, time, the categories. Nature, for us, is the sum of appearances under laws. Now the cloud stands between every man and every appearance, and arranges them for each. It does not merely lie to him here and there. It sets the conditions under which anything can appear to him at all. Give each man his own conditions and you have given him his own nature, which is to say his own world.

Observe also: once a picture can be made of anything, no picture proves anything. Two professors of law have named this the liar's dividend. The guilty man now says of a true recording that it is a fake, and a minister in Spain is reported to have said it of footage of police striking voters. The photograph was once a witness. In the cloud it is one more appearance.

The third was a barefoot old man with a snub nose, whom someone had fetched from a riverbank.

Socrates. It seems also that it does, and that you are to blame, Master, you and all the bookmen. I told Phaedrus of the Egyptian king who refused the gift of letters. Writing, he said, would "create forgetfulness in the learners' souls," because they would trust to marks outside themselves and not remember of themselves. I told Phaedrus too that writing is like painting: its figures look alive, "and yet if you ask them a question they preserve a solemn silence." Your books made the first cloud. They shut each reader in a room with a thing that cannot answer him. The cloud has only finished the work.

McLuhan. We have your objection in writing, I notice.

Socrates. That was Plato's doing. *(Laughter.)*

The fourth was a young woman at the back, a telephone in her hand.

A student. It seems that it does, because the common world is gone, and I can count it. Of the web pages that existed in 2013, 38 percent could no longer be reached ten years later. A quarter of everything put online between 2013 and 2023 had already vanished, and more than half of Wikipedia's pages cite at least one source that no longer exists. On one evening in February 1983, about 106 million Americans watched the last episode of one television programme together. There is no such evening now. Nobody my age has ever had one. If there is no common world, there are many worlds.

The fifth and last was a hairy man girt with a girdle of leather. He had come in by the door and stood beside the empty chair without sitting in it.

Elijah. It seems that it does, and I speak from experience. On Carmel I watched four hundred and fifty prophets of Baal cry from morning until noon and cut themselves with knives, until at last "there was neither voice, nor any to answer, nor any that regarded." That is how your people live now: crying all day into a thing that answers each of them with himself. Afterwards I fled to Horeb and lay in a cave, and when the LORD asked me what I was doing there, I told him the truth as I saw it. "I, even I only, am left." A world of one man. If a prophet can come to that, anyone can.

He did not sit. The master looked at him for a while before turning to the bachelor.

III. On the contrary

McLuhan. On the contrary. Read it, brother.

The bachelor read from a slip of paper, in a soft northern voice.

The bachelor. On the contrary is what the gentleman from Königsberg wrote in 1770, when he took his chair. A world, he said, is not a heap of substances pushed together by an act of thought. It is sub-

stances held together by real bonds, acting on one another. And if there were several first causes, the upshot would be "worlds, not a world." There is one first cause. Therefore there is one world.

Kant. That was before I woke from my dogmatic slumber.

McLuhan. Some men see more clearly asleep. *(Laughter.)* Answer, brother.

The bachelor. I answer, saving the master's better judgment, as a man answers who knows one thing well and little else. I once wrote to a young friend about two clouds. Above him, between him and his God, a cloud of unknowing, on which he was to beat with "a sharp dart of longing love." Beneath him, a cloud of forgetting, which he was to put there of his own will, "betwixt thee and all the creatures that ever be made," so that nothing would come between him and the work of love.

Mark the difference. My friend put his cloud there himself, for love, for a time, and he knew what it covered. Yours was put there by others, for profit, and you do not know that anything lies under it. But neither cloud unmakes the creatures. Forgetting is not destroying. Under my cloud of forgetting, every creature I told my friend to forget was still in its place, held in being by the same God. So I say the cloud does not make many worlds. It makes one world forgotten.

A woman on the second bench, smoking (the beadle had given up), raised her hand.

Arendt. Forgotten is too gentle, brother, and you are confusing two things I took care to keep apart. A man can be isolated and still have a world. The craftsman alone in his shop still makes a table that will stand among other people's things after he is dead. Loneliness is something else. It is "not belonging to the world at all." That is what is happening here.

The world is what lies between us, the way a table stands between the people sitting around it. It relates us and it separates us; it is why two people can look at one thing from two sides. Now imagine a séance at which the table suddenly vanishes. The people sitting opposite one another are no longer separated, but nothing they can touch relates them either. That is your cloud. Not many worlds. No table.

Long before any of this was built, I wrote that what people now have in common is "not the world but the structure of their minds." Your machine is that sentence in silicon. Everyone runs the same application, and no one shares the same world.

The bachelor. I accept the correction. Not a world forgotten: a table hidden.

McLuhan. Hold on to that. We'll come back to it.

IV. Voices from the floor

Before the master could go on, there was noise on the benches. He let it run, as masters did when the hall had more to say than the opponents.

Baruch. Master, I am a scribe and not a disputant, but I have something to the point. My master Jeremiah dictated to me every word the LORD had given him, and the scroll was read before the king. It was winter, and the king sat by a fire on the hearth. "When Jehudi had read three or four leaves, he cut it with the penknife, and cast it into the fire," until the whole roll was burned. That was the first deletion.

Then the word came: "Take thee again another roll." I wrote it all again from his mouth, "and there were added besides unto them many like words." A king can burn only the copy he holds.

And when the city was falling, my master bought a field at Anathoth from his cousin, with the Babylonians at the gate. He told me to put the deeds "in an earthen vessel, that they may continue many days." A field no one could farm, in a country about to be lost. The jar was a promise that fields would be bought and sold in that land again.

An Englishman *(thin, moustached, coughing).* The king with the penknife has new tools. In July 2009 a bookseller reached into thousands of reading machines and took back copies of two books that people had bought and paid for. It refunded the money and said it would not do so again in like cases. The books were *Nineteen Eighty-Four* and *Animal Farm*. *(Laughter, not all of it comfortable.)* I wrote about a memory hole. I did not imagine it would come with free delivery.

Then there are the children's books. In 2023 a publisher revised Roald Dahl for present tastes: words about weight, mental health, violence, gender and race. Augustus Gloop was no longer fat. Readers reported that copies already on their machines had changed without their asking. The outcry was loud enough that the old texts were kept in print, on paper, as a "classic" collection. Paper is the copy the king cannot reach.

Mandelstam *(a small man, his head thrown back).* A poem is a letter in a bottle. A sailor in trouble seals his name and his fate inside and throws it into the sea. Years later someone walking the dunes finds it, opens it and reads it, and it was addressed to him. He is its secret interlocutor. I said this in 1913. Later my wife had to keep my poems in her memory, because paper had become too dangerous. A poem is never written to no one. The bond between the one who throws and the one who finds is made when the bottle leaves his hand. The sea cannot break it; it can only delay it.

Celan. I took the bottle from him. A poem, I told them at Bremen, is sent out in the not always hopeful belief that it may wash up somewhere, "perhaps on a shoreline of the heart." It is on its way toward something open: a you that can be addressed. Even a poem written after what I lived through is headed for someone.

Chesterton *(filling most of a bench).* And they are coming back! Master, people are buying gramophone records again. Last year Americans spent more than a billion dollars on them, for the first time since 1983, the very year of the young lady's television evening. It has grown nineteen years running. A thing you can drop is a thing you own, and a thing you can lend is a thing you share. "Tradition means giving votes to the most obscure of all classes, our ancestors. It is the democracy of the dead." The record shop is a polling station.

The Scotsman *(mildly, from the end of a bench).* I would add only that a printed book is a perfect and just weight. The same page is the same page for every reader, rich or poor. Page seventy-two can be quoted, disputed and checked. A sentence shown to one reader alone can be none of those things.

McLuhan. Thank you, Professor Smith. You have just explained the footnote. *(Laughter.)* Enough. I will determine.

V. The master determines

McLuhan. I answer that the question is wrongly put, which is the usual way with good questions. You ask whether the cloud makes many worlds. It does not. It makes no world. The difference is the whole matter.

First: there cannot be many worlds. The young Kant was right, and I do not think the old Kant ever quite took it back. A world is a web of real bonds, and a web with one source is one web. Worlds in the plural would need first causes in the plural, as many gods as there are subscribers. There is one. Thomas says that as long as a thing has being, God must be present to it, "according to its mode of being." Not once at the start, like a man who winds a clock and leaves the room, but all the time. Every creature on these benches, living or dead, is held in being at this moment by the same hand. No engineer can cut that bond, because no engineer made it. There is no place off that graph for a private world to stand. A soul with no thread back to the source would not be in another world. It would not be.

So when you say *isolated worlds*, you flatter the machine. You credit it with a creator's power, the power to give each of you a cosmos of your own. The Gnostics had a name for an engineer who claims that power: the demiurge. The cloud is no demiurge. It cannot make anything be. It can only hide.

Second: what it hides is the table. Dr Arendt has it. A world, for us, is not only the bonds God holds. It is also the things people make and set between themselves, which relate them and separate them: the altar, the book, the song everyone knows, the picture on the parish-hall wall, the evening when a hundred million people watch the same farewell. That is the furniture of a common world. It is what you can point to and say: *you see it too.* The cloud takes it away one piece at a time, not by burning it but by serving each of us a different piece.

Third: how it does this. For sixty years I have said that the medium is the message, and for sixty years you have asked me what the message is. Here it is. The message of the cloud is not its content. Its content is everything: true news and false, sermons and filth, your grandmother's photographs and a stranger's death. The gentleman from Königsberg is right that the harm goes deeper than false content, but wrong about where it goes. The cloud gives no one private forms of space and time; your minds are made as they always were. What it removes is the common object. Each of you is served alone, adjusted to yourself, and the serving is the message. The same film arrives with a different picture on the box. The same news arrives with a different story attached. The same past arrives with different pages missing. Perhaps nothing is false, and nothing is shared. That is disinformation in the strict sense: not information that is wrong, but information that has lost its form, the form of a thing held in common.

Fourth: the light. Electric light, I wrote, is "a medium without a message." It shows everything and says nothing. The cloud is made of light; it is the purest medium ever built. In 1969 I warned my old friend Maritain that because the electric environment is weightless and everywhere, it tempts us to think the world itself is made of spirit. I called it "a reasonable facsimile of the mystical body, a blatant manifestation of the Anti-Christ." The Prince of this World, I added, is a very good electrical engineer. Everyone connected; no one in communion. Rome said in 1971 that communication is more than the passing of ideas: "At its most profound level it is the giving of self in love." The cloud gives you everything except someone.

Here the determination was interrupted. A voice came from somewhere in the hall, very loud. Afterwards no one could agree where it had come from, and no one ever learned whose it was.

A voice *(fortissimo)*. NOTHING! IT IS ALL MADE OF NOTHING! THEY CUT THE WORLD INTO PIECES TO LEARN IT, AND WHAT IS CUT IS NO LONGER WHAT IT WAS. THEY KNOW ONLY WHAT THEY MEASURE, AND WHAT THEY MEASURE IS NOT BEING! THE LORD CHANCELLOR CUT OFF FORM AND END. HE SAID THE SEARCH FOR ENDS WAS BARREN, LIKE A VIRGIN CONSECRATED TO GOD! AND OUT OF WHAT WAS LEFT THEY SPUN YOUR WIRES AND YOUR GLASS THREADS AND YOUR RADIO WAVES. YOUR CLOUD IS BUILT ON NOTHING. TEAR IT DOWN!

Several of the dead covered their ears. The prophet by the door did not turn his head.

McLuhan *(quietly)*. There is no need to shout. The LORD never does. *(Laughter.)* Half of what you say is true, and you have shouted the wrong half.

The true half: to learn a thing by cutting it is to learn about its parts, never what it is. The physicists measure, and to measure they set form and end aside. That is honest work, as long as they remember what they set aside. The Lord Chancellor did call the search for ends barren. And the things built from that knowledge inherit the bracket: a wire is made for power, and what the power is for is left

open. Put my old saying in Aristotle's words and that is all it means. When the final cause is set aside, the means becomes the end. The medium becomes the message because no one asked what the message was for.

The false half: what they measure is not nothing. Quantity is real; it is the first thing about any body. Copper is not nothing. The current in that wire is not nothing; it carried your shout to the back of the hall. Every electron in it is held in being, this moment, by the same hand that holds you. Call physics the knowledge of nothing and you make the mistake of Galileo's judges backwards. They let theology answer a question of fact; you let it deny one.

If you want the old word, the medium is like prime matter: no form of its own, ready to take any. Augustine called formless matter "near to nothing." Near to nothing is not nothing. Matter never exists without form, and the world never exists without God.

So you need not tear the cloud down. You need to give it back its end. The Church has a formula for that, older than the Lord Chancellor: the word comes to the element, and it becomes a sacrament. Water is only water until the word comes to it, and then it washes souls. Nothing is added to the water's being. Its end is given back.

One more thing. We have two men without names in this hall tonight. One wrote softly, and has been read for six hundred years. You may shout as loud as you like. You will be gone by the next refresh.

No one answered. The master went on.

Fifth: the boy and his father. Forgive me; I wrote a chapter on the comics and I still read them. In 1989 a newspaper strip had a small boy ask his father why old photographs are black and white. The father explained, with a straight face, that the photographs were in colour; it was the world that was black and white, until some time in the 1930s. Then why are old paintings in colour? Artists, said the father, are strange people. And why didn't the old photographs turn colour with everything else? Because they were "color pictures of black and white, remember?"

That strip holds the whole of epistemology in a few panels. The father posits two worlds, a grey one and a coloured one, and the joke works because everyone knows there was only ever one world and two media. We are always tempted to take the limits of the medium for the limits of the world. The opponents did it this morning. Because the cloud shows each of us something different, they conclude that each of us lives somewhere different. It is the father's joke, told without the wink. And mark who kept the colour: the painter. He stood in the world, looked with his own eyes, and made a thing with his hands that other people could stand in front of. The thing survived and carried the colour with it.

Sixth: the reverse. Our grandchildren may make the father's mistake about us. One of the men who built the network warned in 2015 that this century could become a digital dark age: everything recor-

ded, little kept. Our descendants will not think we lived in black and white. They will think we lived in a blank, unless something survives.

Seventh: what survives. The book, the record, the cassette, the painting, the deed in the jar. You think these things survive because they are old-fashioned. They survive because they are things. They have weight. They are the same for everyone who holds them. They can be lent, inherited, dropped, carried next door, and burned only one copy at a time. Every medium pushed to its limit flips into its opposite and retrieves something an earlier one had made obsolete; the cloud, pushed far enough, has made people hungry for something they can hold. These things are not nostalgia. They are the legs of the table showing through the cloud. The table was never taken away; it was hidden. And the bonds beneath it, the ones that hold us in being and the ones we made with each other, were never cut. Only clouded.

That is my determination. Not many worlds. Above the cloud, for now, no world; beneath it, the one world, whole, with all its bonds holding.

VI. Replies to the objections

McLuhan. To the first, Herr Leibniz: finish your sentence.

Leibniz *(smiling, and reading).* "...so, because of the infinite number of simple substances, there are a similar infinite number of universes which are, nevertheless, only the aspects of a single one as seen from the special point of view of each monad."

McLuhan. *Only the aspects of a single one.* You never believed in many worlds. You believed in many views of one, even while you said there were no windows. And God established your harmony before the world began, so that every mirror would agree. The harmony of the cloud is arranged after the fact, by a company, so that no two mirrors need agree. Yours was pre-established. Theirs is post-arranged, and its purpose is not to harmonize anything. It is to keep each of you from noticing the others.

To the second, Herr Professor: you have answered yourself twice. Once in 1770, when you said a world needs real bonds and one source. And again in 1786, when you were told that a tyrant may take away the freedom to speak and write but never the freedom to think. You replied that we would think very little, and very badly, if we did not think in company with others, to whom we tell our thoughts and who tell us theirs; so whatever takes away public communication takes away the freedom to think as well. The cloud has found how to do this without forbidding anything. It lets everyone speak, and arranges that no two people hear the same thing.

As for the liar's dividend, I grant it. A picture anyone can make is no witness. But a witness was never a picture. A witness is a person who was there and will put his name to it. That is why the painting outlasts the photograph as testimony: there was always somebody behind it.

To the third, Socrates: the king was right that letters breed forgetfulness. Every extension is also an amputation; I have said so all my life. But you were wrong that a book is silent the way a painting is silent. It is silent the way a letter in a bottle is silent, until someone opens it. Then it answers him, and gives him the same answer it gave everyone else, which is exactly the virtue you mistook for a fault. The cloud is never silent. It answers every man at once, each with himself. And you will notice that your objection reached this hall because a young man wrote it down. *(Laughter.)*

To the fourth, my young friend with the telephone: everything you counted is true, and none of it proves many worlds. A page that can no longer be reached is not in another world. It is gone, and the bonds it carried must be carried by something else. Your evening of 1983 is gone too, and it was partly a manufactured thing: three networks chose what a nation would remember. But a table can be set again. It is set every Sunday at Mass and every spring at Passover, and in every library and every shop that sells records. Go and sit at one.

To the fifth I will not reply. The prophet's objection belongs to the prophet.

The hall turned. The man in the leather girdle was still standing beside the empty chair.

Elijah. I said, "I, even I only, am left," and I was answered. First the LORD told me to stand upon the mount. "And, behold, the LORD passed by, and a great and strong wind rent the mountains, and brake in pieces the rocks before the LORD; but the LORD was not in the wind: and after the wind an earthquake; but the LORD was not in the earthquake: And after the earthquake a fire; but the LORD was not in the fire: and after the fire a still small voice." Your cloud is wind, earthquake and fire, all day long, in every hand. The LORD chose the thinnest medium there is. Those who know my tongue say it was the sound of a thin silence.

Then he answered my complaint. "Yet I have left me seven thousand in Israel, all the knees which have not bowed unto Baal." Seven thousand, and I had not seen one of them. I thought I was a world of one man. I was one of seven thousand bonds I could not see. That is the error of every one of you who says the cloud has sealed him in a world of his own. Your seven thousand are there.

And before that, on Carmel, while the others cried to a god who did not answer, I did not argue with them. I "repaired the altar of the LORD that was broken down." I took "twelve stones, according to the number of the tribes of the sons of Jacob." The kingdom was split in two, and I built one altar out of all twelve tribes, from stones a man could put his hands on. That is what your books and records are, Master. Twelve stones.

The last word of the old prophets is about me. "Behold, I will send you Elijah the prophet before the coming of the great and dreadful day of the LORD: And he shall turn the heart of the fathers to the children, and the heart of the children to their fathers." That is my commission: the bond between the generations, the common memory. It is why the Jews still pour a cup for me at Passover and open the door. It is why your beadle set this chair.

He sat down in it.

VII. The Light pitched his tent

It was late. The master folded his newspaper and spoke more quietly.

McLuhan. One word more, and then the bell. The prophet in that chair has stood in light before, on a mountain. He and Moses talked with our Lord while "his face did shine as the sun, and his raiment was white as the light." Peter, who never knew what to say and always said it, offered to put up tents: "Let us make here three tabernacles; one for thee, and one for Moses, and one for Elias." And "a bright cloud overshadowed them." Mark that. Not a cloud of forgetting, nor even of unknowing, but a bright cloud with a voice in it: "This is my beloved Son, in whom I am well pleased; hear ye him." And when they lifted up their eyes, "they saw no man, save Jesus only."

Peter wanted to put the light in a tent. He was too late; the Light had already pitched his own. Saint John tells it: "the Word was made flesh, and dwelt among us." His word for *dwelt* means that he pitched his tent. The Light of the world did not become a cloud, weightless, everywhere and nowhere, adjusted to each. He became one man, in one place, at one time, with a face you could see and hands you could touch. He is the only medium in history of which it can be said that there is "no distance or separation between the medium and the message." He is what he says.

And he left himself to us the same way. Not as information: as a thing. Bread, under which he is present truly, really and substantially, in every host and every fragment, for as long as the species last. We keep him in a tabernacle, which is to say a tent, on street after street of this city, behind a small red lamp. If you want to know where the common world is still kept whole, go and look at that lamp. It is the last table, and the first.

Saint Paul knew the pattern. "For God, who commanded the light to shine out of darkness, hath shined in our hearts … But we have this treasure in earthen vessels." Jeremiah's jar. Elijah's stones. Your record and your book. The deed to a field in a country about to be lost, sealed "that they may continue many days."

And mark what became of the prophet's media. On Horeb the LORD was not in the wind, and not in the fire. But when the prophet's work was done, "there appeared a chariot of fire, and horses of fire … and Elijah went up by a whirlwind into heaven." The same wind and the same fire that were

emptied on the mountain were taken up to carry a man home. His cloak, a piece of cloth, parted the Jordan in his disciple's hands, because the disciple asked as he struck the water, "Where is the LORD God of Elijah?" Nothing was added to the wind, the fire or the cloth. They were given back their end.

So it will be with your cloud. It will not be blown away; it will be taken up. When our Lord ascended, "a cloud received him out of their sight," and Saint Paul says we shall be "caught up together with them in the clouds." The cloud of forgetting will be a bright cloud again, with a voice in it.

So take courage. Moses wrote that in the beginning "the earth was without form, and void; and darkness was upon the face of the deep," until God said, "Let there be light." Saint John opens his gospel with Moses's first words, In the beginning, and writes that "the light shineth in darkness; and the darkness comprehended it not." His Greek word is a good one, and it will not choose for you. The darkness did not grasp the light: it did not understand it, and it did not overpower it. The cloud does not understand the light, and it cannot put it out. It is made of a borrowed brightness it will never comprehend. The Light that pitched his tent among us is not in the wind, nor in the earthquake, nor in the fire. He is in the still small voice, and in the bread, and in the stones you can put your hands on. The darkness shall not overcome it.

He stood. The bachelor bowed. The prophet in the chair by the door did not move.

McLuhan. The disputation is ended. *Ite, disputatio est.* Go home and lend someone a book.

The bell rang. As it rang, someone near the back (it was never established who) got up, opened the door, and held it, as though someone else were expected.

Coda: the music

The disputation has a musical twin: a short piece for Strudel, a free live-coding environment that makes music in a web browser. Like the hall, it is built from its end. The last thing you hear is the opening of the Ascension introit, *Viri Galilaei* ("Men of Galilee, why stand you looking up to heaven?"). The first thing you hear is the same phrase backwards, in fog, as in Machaut's rondeau *Ma fin est mon commencement*. Its pulse is the first message ever sent by electric telegraph, "What hath God wrought" (Numbers 23:23), tapped in Morse in May 1844. By the end that rhythm carries the chant: the fourteenth century's isorhythm, a rhythm and a melody of different lengths turning against each other. The medium is given back its word.

1. **I. The cloud** (0:00): the introit's opening, backwards and filtered, with a telegraph far off.
2. **II. The hall** (0:24): a fugato on the introit: subject, answer over the held reciting tone, then the bass.
3. **III. The Voice** (0:56): two seconds of noise, fortissimo, and two of silence.
4. **IV. Horeb** (1:04): wind, earthquake and fire, each cut off: the LORD was not in them.

5. **V. A still small voice** (1:28): “Lumen Christi” on a pure sine tone, three times, each a step higher, answered “Deo gratias”.
6. **VI. A bright cloud** (1:52): “Rorate caeli,” with the telegraph’s rhythm now carrying the chant on the celesta.
7. **VII. Taken up** (2:24): the introit rising an octave at each entry, the bells ringing it in Morse.
8. **VIII. My end is my beginning** (2:48): the opening phrase, forward and clear, and an open fifth on G.

How to play it yourself

1. Open the piece in Strudel. The link carries the whole score; there is nothing to install or sign up for. A recent desktop browser works best, and headphones help.
2. Press **play** at the top of the editor, or click in the code and press **Ctrl+Enter** (Alt+Enter also works). **Ctrl+.** stops it.
3. The choir, organ and bells are sampled instruments that load the first time they sound, so the first run can stutter. If it does, stop and play again.
4. Set your volume during the opening and leave it there. One passage, about a minute in, is meant to be very loud for two seconds.
5. To play it live, change it while it runs. The code is the score: edit a line and press Ctrl+Enter, and Strudel takes up the new version without stopping. Try another tempo in `setcpm(30)`, another message in `morse('WHAT HATH GOD WROUGHT')`, or remove a section from the `arrange(…)` at the bottom.
6. If the link ever stops working, download the source (or copy it from below), paste it into the editor at strudel.cc, and press play. The PDF of this page prints the whole source on paper.

The chant pitches are taken from GregoBase transcriptions of the *Graduale* and the 1961 Missal; the melodies are public domain. Strudel is free software by its own authors. The piece itself, the code below, is dedicated to the public domain with this essay.

The source code

```
// WHETHER THE CLOUD MAKES MANY WORLDS
// A musical analogue to the disputation of the same name.
// For Strudel (https://strudel.cc). Paste this in and press play.
// About three minutes and twenty seconds. It gets very loud once, briefly.
//
// Dedicated to the public domain (CC0 1.0). The chant melodies are public
// domain; the pitches are taken from GregoBase transcriptions (CC0).
//
// The form is built from its end. The last thing you hear is the opening of
// the Ascension introit, "Viri Galilaei". The first thing you hear is that
```

```
// same phrase backwards, in fog: Machaut's "Ma fin est mon commencement".

setcpm(30) // one cycle = two seconds

// ---------- the sources ----------
// Durations in units: a syllable on one note takes 2, notes of a neume take 1,
// the end of a phrase takes more. "~" is a breath.

// "Viri Galilaei, quid admiramini" (Acts 1:11), introit for the Ascension, mode
7
const VIRI = 'g3@2 g3@2 g3 c4 b3 c4@2 d4 c4 e4 d4 d4@3'
// the same phrase, end first
const IRIV = 'd4@3 d4 e4 c4 d4 c4@2 b3 c4 g3 g3@2 g3@2'
// the answer, a fifth above
const ANSWER = 'd4@2 d4@2 d4 g4 f4 g4@2 a4 g4 b4 a4 a4@3'
// mode 7 recites on D: the tenor holds it while the answer sings
const TENOR = 'd4@12 c4@4'
// two voices above the bass entry
const ALTO = 'd4@2 d4@2 d4 e4 d4 c4@2 f4 g4 g4 a4 a4@3'
const TREBLE = 'b4@2 b4@2 b4 c5 d5 e5@2 d5 e5 g5 f5 f5@3'

// "Rorate caeli desuper, et nubes pluant justum" (Isaiah 45:8), Advent, mode 1
const RORATE = 'f3@2 g3 a3 a3@3 d4@2 c4@2 c4 bb3 a3@2 a3@2 g3@2 ~ ' +
               'a3@2 bb3@2 a3@2 g3 f3 e3@2 f3 e3 d3@3 ~@5'

// "Lumen Christi" / "Deo gratias", the Easter Vigil: do do do la, raised each
time
const lumen = (doh, la) => doh + '@2 ' + doh + '@2 ' + doh + '@2 ' + la + '@3 ~'
const gratias = (doh, la) => doh + '@2 ' + doh + '@2 ' + doh + '@2 ' + la + '@2
' + la + '@3 ~'

// ---------- the medium ----------
// The first message sent by electric telegraph, 24 May 1844 (Numbers 23:23).
// Dot 1 unit, dash 3; 1 between signs, 3 between letters, 7 between words.
const CODE = { W: '.--', H: '....', A: '.-', T: '-', G: '--.', O: '---', D:
'-..', R: '.-.', U: '..-' }
const morse = (text) => text.split(' ').map((word) =>
  word.split('').map((ch) =>
    CODE[ch].split('').map((sign) => (sign === '.' ? 'x' : 'x@3')).join(' ~ ')
  ).join(' ~@3 ')
).join(' ~@7 ')
const TALEA = morse('WHAT HATH GOD WROUGHT') // 189 units, 50 strokes
const SPAN = 189 / 16 // one unit = a sixteenth of a cycle

// Isorhythm, the fourteenth century's own device: a rhythm (talea) carries a
// melody (color), one pitch per stroke. The medium given back its word.
const pitches = (phrase) => phrase.split(' ').filter((t) => t && t[0] !==
'~').map((t) => t.split('@')[0])
const iso = (talea, color) => { let i = 0; return talea.replace(/x/g, () =>
color[i++ % color.length]) }
```

```
const telegraph = (pitch) => note(mini(iso(TALEA, [pitch]))).slow(SPAN)
const ringing = (phrase, up) => note(mini(iso(TALEA,
pitches(phrase)))).transpose(up).slow(SPAN)

// a phrase sung over n cycles
const sing = (phrase, n) => note(mini(phrase)).slow(n)

// ---------- I. The cloud (12 cycles) ----------
// The ending, backwards, through fog. A telegraph somewhere far off.
const cloud = stack(
  arrange([8, sing(IRIV, 8)], [4, silence])
    .s('gm_voice_oohs').lpf(700).room(0.9).gain(0.35),
  s('pink').attack(0.8).release(0.8).lpf(600).gain(0.06),
  arrange([4, silence], [8, telegraph('a5')])
    .s('sine').hpf(1500).delay(0.5).delaytime(0.375).delayfeedback(0.5).room(0.8
).gain(0.03),
  arrange([8, silence], [4, note("[g2,d3]")])
    .s('gm_church_organ').attack(2).gain(0.12),
)

// ---------- II. The hall (16 cycles) ----------
// A fugato on the introit: the subject over a drone, the answer over the held
// tenor, then the bass with two voices above it.
const hall = stack(
  arrange([4, note("[g2,d3]")], [12, silence]).s('gm_church_organ').gain(0.15),
  arrange([4, sing(VIRI, 4)], [4, sing(TENOR, 4)], [4, sing(ALTO, 4)], [4,
sing('c5 b4', 4)])
    .s('gm_choir_aahs').room(0.6).gain(0.4),
  arrange([4, silence], [4, sing(ANSWER, 4)], [4, sing(TREBLE, 4)], [4, sing('e5
d5', 4)])
    .s('gm_choir_aahs').room(0.6).gain(0.32),
  arrange([8, silence], [4, sing(VIRI, 4).transpose(-12)], [4, sing('c3 g2',
4)])
    .s('gm_church_organ').gain(0.3),
  arrange([8, silence], [4, telegraph('a5').gain(0.05)], [4,
telegraph('a5').gain(0.1)])
    .s('sine').room(0.3),
)

// ---------- III. The Voice (4 cycles) ----------
// No name. Fortissimo. Then nothing at all.
const voice = arrange(
  [2, stack(
    s('white').gain(0.6),
    s('crackle').density(0.6).gain(0.45),
    note("[g2,ab2,a2,bb2,b2,c3,db3]").s('sawtooth').distort(2).gain(0.1),
  )],
  [2, silence],
)
```

```
// ---------- IV. Horeb (12 cycles) ----------
// "But the LORD was not in the wind ... not in the earthquake ... not in the
fire."
const horeb = arrange(
  [3, s('wind').n("<0 4 7>").gain(0.4).room(0.4)],
  [1, silence],
  [3, stack(s('brown').lpf(140).gain(0.4),
note('g1').s('sawtooth').lpf(90).gain(0.3))],
  [1, silence],
  [3, s('crackle').density(0.25).hpf(800).gain(0.35)],
  [1, silence],
)

// ---------- V. A still small voice (12 cycles) ----------
// A pure tone, the thinnest medium there is. The people answer, three times,
// each time a step higher.
const still = stack(
  arrange(
    [2, sing(lumen('d4', 'b3'), 2)], [2, silence],
    [2, sing(lumen('e4', 'c#4'), 2)], [2, silence],
    [2, sing(lumen('f4', 'd4'), 2)], [2, silence],
  ).s('sine').attack(0.05).release(0.6).room(0.7).gain(0.22),
  arrange(
    [2, silence], [2, sing(gratias('d4', 'b3'), 2)],
    [2, silence], [2, sing(gratias('e4', 'c#4'), 2)],
    [2, silence], [2, sing(gratias('f4', 'd4'), 2)],
  ).s('gm_voice_oohs').room(0.7).gain(0.22),
)

// ---------- VI. A bright cloud (16 cycles) ----------
// "Drop down, ye heavens, and let the clouds rain down the Just One."
// The telegraph's rhythm now carries the chant.
const bright = stack(
  note("[d2,a2]").s('gm_church_organ').gain(0.14),
  sing(RORATE, 8).s('gm_choir_aahs').room(0.6).gain(0.38),
  arrange([4, silence], [12, ringing(RORATE, 24)])
    .s('gm_celesta').room(0.8).gain(0.18),
  note("[d4,a4]").s('gm_pad_halo').lpf(saw.range(500, 3000).slow(16)).gain(0.1),
)

// ---------- VII. Taken up (12 cycles) ----------
// "There appeared a chariot of fire ... and Elijah went up by a whirlwind."
// The introit rises an octave at each entry; the bells carry it on the
telegraph.
const takenUp = stack(
  note("[g1,g2]").s('gm_church_organ').gain(0.18),
  arrange([4, sing(VIRI, 4).transpose(-12)], [8, note('d3')])
    .s('gm_church_organ').gain(0.28),
  arrange([4, silence], [4, sing(VIRI, 4)], [4, note('d4')])
    .s('gm_choir_aahs').room(0.6).gain(0.36),
```

```
  arrange([8, silence], [4, sing(VIRI, 4).transpose(12)])
    .s('gm_choir_aahs').room(0.6).gain(0.4),
  ringing(VIRI, 12).s('gm_tubular_bells').room(0.7).gain(0.16),
)

// ---------- VIII. My end is my beginning (16 cycles) ----------
// The opening phrase, forward and clear. Then the open fifth.
const end = stack(
  arrange([8, note("[g2,d3]")], [8, silence]).s('gm_church_organ').gain(0.12),
  arrange([8, sing(VIRI, 8)], [8,
silence]).s('gm_voice_oohs').room(0.7).gain(0.38),
  arrange([8, silence], [8, note("[g2,d3,g3,d4]")])
    .s('gm_church_organ').gain("<0.3 0.27 0.24 0.2 0.16 0.12 0.08 0.04>"),
)

arrange(
  [12, cloud],
  [16, hall],
  [4, voice],
  [12, horeb],
  [12, still],
  [16, bright],
  [12, takenUp],
  [16, end],
)
```

A note on what is claimed

Every factual claim below was checked against the source named; documents and measurements settle a question, and reports leave it open. The verdicts ("post-arranged harmony," "disinformation in the strict sense," "no world") are the speakers' judgments, not findings.

- 38% of web pages from 2013 unreachable by October 2023; a quarter of all pages from 2013–2023 gone; 54% of Wikipedia pages cite at least one dead link — *Documented; Pew Research Center, May 2024*
- *M*A*S*H* finale, 28 February 1983: about 106 million US viewers — *Documented; Wikipedia*
- Netflix chooses a title's artwork per member from viewing history — *Documented (Netflix's own disclosure, Dec 2017); Dezeen*
- "Liar's dividend" coined by Chesney and Citron, 2019 — *Documented; Tech Policy Press*
- A Spanish minister called footage of police striking Catalan voters fake — *Reported; Tech Policy Press*
- Amazon removed bought copies of *1984* and *Animal Farm* from Kindles, July 2009; refunded; said it would not do so in like cases — *Documented; History of Information, after the NYT*
- Puffin revised Dahl's texts in 2023 (weight, mental health, violence, gender, race); the old texts kept in print as a classic collection — *Documented; The Bookseller*

- Copies already on readers' devices changed automatically — *Reported, not confirmed by the publisher; Reclaim The Net, citing The Times*
- US vinyl revenue over $1 billion in 2025, first time since 1983; 19th straight year of growth — *Documented; RIAA, via Music Week*
- Warning of a "digital Dark Age," February 2015 — *Documented; NPR*
- Nadezhda Mandelstam kept her husband's poems by memory — *Well attested in her memoirs; not fetched for this note*
- First electric telegraph message, Washington to Baltimore, 24 May 1844, from Numbers 23:23 — *Documented; Library of Congress*

Words quoted as the speakers' own. Leibniz, *Monadology* §§7, 57 (trans. George Montgomery). Plato, *Phaedrus* 275a, 275d (trans. Benjamin Jowett). Kant, *Inaugural Dissertation* (1770) §21 (trans. Jonathan Bennett), and "dogmatic slumber" from the *Prolegomena* (1783); the 1786 essay "What Does It Mean to Orient Oneself in Thinking?" is paraphrased. *The Cloud of Unknowing*, chs. 5–6. Arendt, *The Origins of Totalitarianism* (1951) and *The Human Condition* (1958); the table and the séance are paraphrased. Aquinas, *Summa theologiae* I q.8 a.1. Augustine, *Confessions* XII.7 (*prope nihil*, "near to nothing") and *Tractates on John* 80.3 (*accedit verbum ad elementum, et fit sacramentum*), translated. Bacon on final causes, *De augmentis scientiarum* III.5, paraphrased. Mandelstam's "On the Interlocutor" (1913) is paraphrased. Celan, Bremen speech (1958), trans. Rosmarie Waldrop. Chesterton, *Orthodoxy* (1908), ch. 4. McLuhan, *Understanding Media* (1964), ch. 1; letter to Jacques Maritain, 6 May 1969 (*Letters of Marshall McLuhan*, 1987, pp. 369–71); *The Medium and the Light* (1999), p. 103. Pontifical Commission for Social Communications, *Communio et Progressio* (1971) §11. Scripture is the King James Version: Genesis 1:2–3; 1 Kings 18–19; 2 Kings 1:8 and 2:11–14; Jeremiah 32 and 36; Malachi 4:5–6; Matthew 17:1–8; John 1:5, 14; Acts 1:9; 2 Corinthians 4:6–7; 1 Thessalonians 4:17. The comic strip (*Calvin and Hobbes*, 29 October 1989) is paraphrased, with one line quoted. The loud voice has no name and no source.

Still to check against print: Jowett's wording; the Aquinas translation; the 1786 passage (Ak 8:144); the page of Arendt's séance; McLuhan's letter in the 1987 *Letters*.

What the other side argues. The best large study of the 2020 US election found that most of what Facebook users see comes from like-minded sources, but that cutting that exposure by a third changed none of eight measures of polarization or belief. That tells against "isolated worlds," as the hall concludes; it does not show "no world," which is the master's verdict, not a finding. The common culture of 1983 was made by three networks and left many people out, and its fragmentation began with cable, before the internet. Paper burns too, and digital archives such as the Internet Archive keep a great deal of what the web loses; much of what vanished was never meant to last. The vinyl revival is partly a collectors' market. And Socrates' complaint about writing has never been fully answered: every medium that remembers for us also teaches us to forget.

SOURCES Pew Research Center, "When Online Content Disappears" (2024), "Goodbye, Farewell and Amen" (Wikipedia), Dezeen on Netflix's personalized artwork (2017), Tech Policy Press on the liar's dividend (2025), History of Information on Amazon's deletion of Orwell (2009), The Bookseller on the Roald Dahl Classic Collection, Reclaim The Net on Dahl e-books (2023), Music Week on the RIAA's 2025 figures, NPR on the "digital Dark Age" warning (2015), Nyhan et al., "Like-minded sources on Facebook are prevalent but not polarizing" (Nature, 2023), Brett Robinson, "The Patron Saint of Media Studies," Church Life Journal (2018), America on McLuhan's letter to Maritain, Thomas Farrell, "Contextualizing Marshall McLuhan" (the letter's date and pages), A dictionary of McLuhan's quotations (The Medium and the Light, p. 103), McLuhan, "The Medium Is the Message", Communio et Progressio (1971), The Cloud of Unknowing, ch. 5 (CCEL), Kant, Inaugural Dissertation (trans. Bennett), Leibniz, Monadology §57 (trans. Montgomery), Mandelstam, "The Secret Addressee" (Cerise Press), Celan's Bremen speech (Poetry Unbound), Arendt on isolation and loneliness (The Marginalian), Arendt on the common world and common sense (Christian Study Center), Calvin and Hobbes, 29 October 1989 (WIST), Library of Congress, "What Hath God Wrought?".

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